Problem Set: 3
Computer Networks and the Internet: Transport Layer

1 Essay Questions

1. Is it possible for an application to enjoy reliable data transfer even when the application runs over UDP? If so, how?

2. Suppose that a Web server runs in Host C on port 80. Suppose this Web server uses persistent connections, and is currently receiving requests from two different Hosts, A and B. Are all of the requests being sent through the same socket at Host C? If they are being passed through different sockets, do both of the sockets have port 80? Discuss and explain.

3. In rdt protocols, why did we need to introduce sequence numbers?

4. Which protocol –Go-Back-N or Selective-Repeat makes more **efficient** use of network bandwidth? Why?

2 Problems

5. (a) Suppose you have the following 2 bytes: 01011100 and 01100101. What is the 1s complement of the sum of these 2 bytes?

   (b) Suppose you have the following 2 bytes: 11011010 and 01100101. What is the 1s complement of the sum of these 2 bytes?

   (c) For the bytes in part (a), give an example where one bit is flipped in each of the 2 bytes and yet the 1s complement doesn't change.

6. Consider the Go-Back-N protocol, and suppose that base of the senders window is X. Is it possible for the sender to receive an ACK for a packet that has a smaller sequence number than X. If so, sketch out a timeline-diagram showing sender and receiver messages sent and received that shows how this is possible. If not, explain why this can never happen.
   Hint: The base of the senders window is packet with the least sequence number that is transmitted and waiting to be acknowledged.

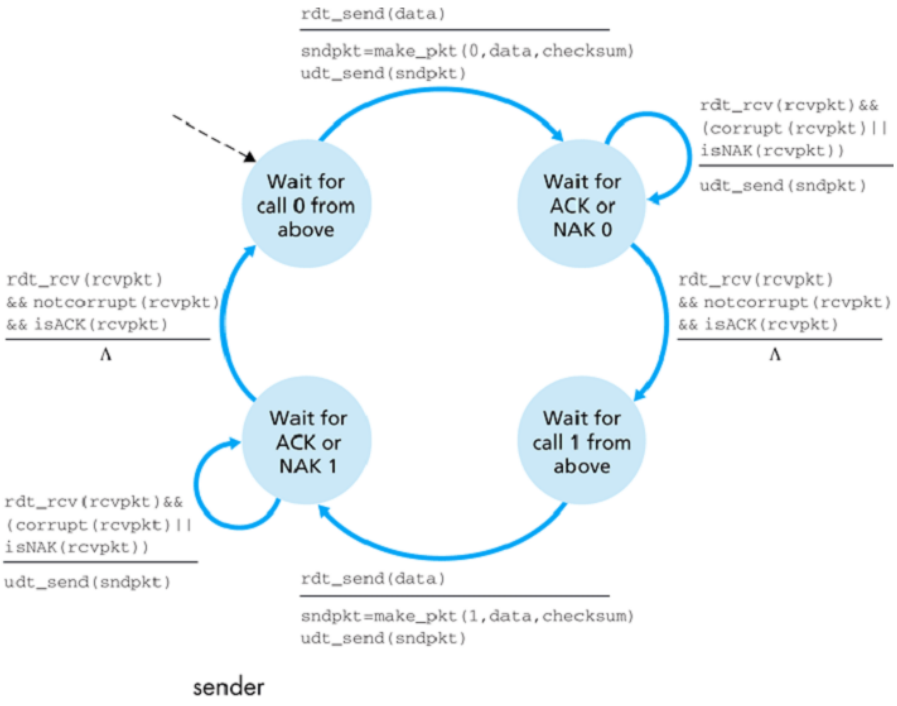
7. Consider the GBN protocol with a sender window size of 4 and a sequence number range of 1,024. Suppose that at time t, the next in-order packet that the receiver is expecting has a sequence number of k. Assume that the medium does not reorder messages. Answer the following questions:

   (a) What are the possible sets of sequence numbers inside the sender's window at time t? Justify your answer.

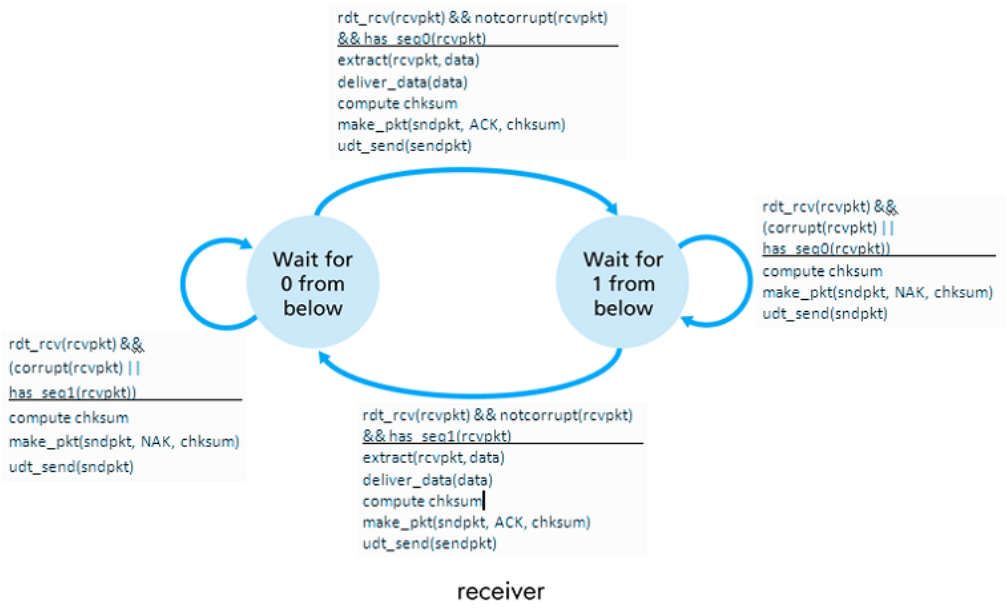
(b) What are all possible values of the ACK field in all possible messages currently propagating back to the sender at time t? Justify your answer.

8. Consider a reliable data transfer protocol that uses only negative acknowledgments. Suppose the sender sends data infrequently. Would a NAK-only protocol be preferable to a protocol that uses ACKs? Why? Now suppose the sender has a lot of data to send and the end-to-end connection experiences few losses. In this second case, would a NAK-only protocol be preferable to a protocol that uses ACKs? Why?

9. Consider the sender and receiver below, show a scenario that they enter into a deadlock state.



(a) Sender Side



(b) Receiver Side

3 Notes

- It's encouraged that you take a look at the Homework Problems for chapter 3, from the kurose/Ross reference.

Good Luck